Exercices : Produits Cartésiens

Maël

3 mars 2026

Énoncé

Soient les ensembles :

$$E = \{x, y\} \qquad F = \{1, 2, 3\} \qquad G = \{a, b\}$$

Leurs cardinaux respectifs sont : $|E| = 2$, $|F| = 3$, $|G| = 2$.

Exercice 1

$$\begin{aligned}
E \times F &= \{(x,1),(x,2),(x,3),(y,1),(y,2),(y,3)\}\\
F \times E &= \{(1,x),(1,y),(2,x),(2,y),(3,x),(3,y)\}\\
E \times G &= \{(x,a),(x,b),(y,a),(y,b)\}\\
G \times E &= \{(a,x),(a,y),(b,x),(b,y)\}\\
F \times G &= \{(1,a),(1,b),(2,a),(2,b),(3,a),(3,b)\}\\
G \times F &= \{(a,1),(a,2),(a,3),(b,1),(b,2),(b,3)\}
\end{aligned}$$

Exercice 2

$$\begin{aligned}
E \times G \times F = \{&(x,a,1),(x,a,2),(x,a,3),(x,b,1),(x,b,2),(x,b,3),\\
&(y,a,1),(y,a,2),(y,a,3),(y,b,1),(y,b,2),(y,b,3)\}
\end{aligned}$$

$$\begin{aligned}
F \times G \times E = \{&(1,a,x),(1,a,y),(1,b,x),(1,b,y),\\
&(2,a,x),(2,a,y),(2,b,x),(2,b,y),\\
&(3,a,x),(3,a,y),(3,b,x),(3,b,y)\}
\end{aligned}$$

Exercice 3

$$E^3 = \{(x,x,x),(x,x,y),(x,y,x),(x,y,y),(y,x,x),(y,x,y),(y,y,x),(y,y,y)\}$$

$$G^3 = \{(a,a,a),(a,a,b),(a,b,a),(a,b,b),(b,a,a),(b,a,b),(b,b,a),(b,b,b)\}$$

$$\begin{aligned} F^3 = \{&(1,1,1),(1,1,2),(1,1,3),(1,2,1),(1,2,2),(1,2,3),(1,3,1),(1,3,2),(1,3,3),\\ &(2,1,1),(2,1,2),(2,1,3),(2,2,1),(2,2,2),(2,2,3),(2,3,1),(2,3,2),(2,3,3),\\ &(3,1,1),(3,1,2),(3,1,3),(3,2,1),(3,2,2),(3,2,3),(3,3,1),(3,3,2),(3,3,3)\} \end{aligned}$$

Exercice 4

— $|E^2 \times F \times G| = |E|^2 \times |F| \times |G| = 2^2 \times 3 \times 2 = 4 \times 3 \times 2 = \mathbf{24}$
— $|E \times F^2 \times G^2| = |E| \times |F|^2 \times |G|^2 = 2 \times 3^2 \times 2^2 = 2 \times 9 \times 4 = \mathbf{72}$